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Exploring outcomes between single and dual caregiver households participating in a parenting and family strengthening intervention in Rwanda: the case of Sugira Muryango

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Abstract

Introduction Early childhood is a critical period for development, yet families in low-resource settings often face barriers that compromise effective caregiving. Research evidence shows that parenting interventions can improve child growth and development. Given the unique challenges faced by single-headed households, this study examined whether caregiver household type modified the effects of Sugira Muryango, an evidence-based home-visiting program in Rwanda. Specifically, it assessed whether changes in child growth and development, dietary diversity, and healthcare-seeking behaviors over time differed between children from single versus dual caregiver households in the Sugira Muryango cluster randomized controlled trial (cRCT).

Methods A total of 1,049 households were enrolled in the Sugira Muryango cRCT, with 508 households assigned to the control group and 541 to the intervention group. The analytic sample included 959 children from 959 households. Of these, 498 were from single caregiver households and 461 from dual caregiver households. Baseline and follow-up data collected 12 months post-intervention were analyzed. Linear mixed-effects models (LMMs) were used to analyze continuous outcomes, accounting for children nested within households and clusters. Models included fixed effects for timepoint, treatment group, caregiver household type (single vs. dual), and their three-way interaction (timepoint × treatment group × caregiver household type). Random intercepts for households and clusters were included to account for the hierarchical data structure. Mixed-effects logistic regression models were used for binary outcomes.

Results There was no evidence of effect modification by caregiver household type on any outcome. For child growth indicators, three-way interaction terms were not significant for weight-for-age z-score (WAZ) ($\beta = 0.081$, 95% CI: -0.077 to 0.239), height-for-age z-score (HAZ) ($\beta = 0.013$, 95% CI: -0.180 to 0.207), or weight-for-height z-score (WHZ) ($\beta = 0.129$, 95% CI: -0.083 to 0.341). Dietary diversity also showed no differential effect by caregiver household type

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($\beta = 0.256$, 95% CI: -0.175 to 0.686). For the Ages and Stages Questionnaire developmental domains, the three-way interaction terms were not significant: Communication ($\beta = -0.272$, 95% CI: -0.779 to 0.235), Gross Motor ($\beta = 0.107$, 95% CI: -0.571 to 0.785), Fine Motor ($\beta = 0.129$, 95% CI: -0.331 to 0.590), Problem-Solving ($\beta = 0.101$, 95% CI: -0.372 to 0.574), and Personal-Social ($\beta = 0.090$, 95% CI: -0.370 to 0.550). No significant modification effects were observed for care-seeking behaviors for diarrhea ($\beta = 2.03$, 95% CI: -0.92 to 4.97) or fever/cough ($\beta = 1.22$, 95% CI: -0.13 to 2.56).

Conclusion The Sugira Muryango intervention yielded similar outcomes across caregiver household types. Given its inclusive design and potential to deliver equitable benefits across diverse households settings in low-resource contexts, it offers a promising model for community-based interventions with important implications for scaling up similar programs.

Keywords Sugira muyango, Family strengthening, Prevention, Early childhood development (ECD), Family configuration, Nurturing care

Introduction

Early childhood is a critical period that shapes lifelong physical, cognitive, and social-emotional development [1]. A nurturing family environment, characterized by stable and responsive caregiving, adequate nutrition, good health, opportunities for early learning, and a safe and secure environment, is essential for children's well-being and positive development [2–6]. However, families in resource-limited settings often face significant challenges in providing these foundational needs due to poverty and systemic barriers such as limited access to quality health and education services, and inadequate social protection systems [7–11]. These structural barriers can undermine family stability, disrupt parenting practices, and negatively influence child developmental outcomes [1, 12, 13]. Additionally, poverty-related stress may limit caregiving capacity, while exposure to violence or neglect further heightens risks of poor health, learning difficulties, and long-term socio-economic disadvantages [14, 15].

In particular, children living in single caregiver households may be at elevated risk due to increased financial, emotional, and time-related burdens placed on their caregivers [16–19]. When combined with limited social support, these factors can result in reduced caregiving quality, directly affecting child health, nutrition, and development compared with those in dual caregiver households [17, 20, 21]. Globally, 6.8% of children live in a single caregiver household [22], yet in Rwanda, this figure is much higher: in 2015, 22% of children lived with only their mother and 2% with only their father [23]. At the same time, over one-third of the Rwandan population lives below the poverty line [24], compounding vulnerabilities among these families.

Early childhood development (ECD) interventions, particularly those using home-visiting approaches have proven effective in promoting positive parenting, building resilience, enhancing child development outcomes,

and supporting early learning [25–30]. However, most ECD programs do not explicitly examine whether household caregiving structure influences responsiveness to such interventions, despite the distinct needs and constraints faced by single caregivers.

In Rwanda, the Vision 2020 Umurenge Program (VUP), which provides direct or conditional cash transfers (based on completion of labor work) to very poor households was introduced to accelerate poverty reduction and enhance social protection [31]. Labor work includes traditional public works, which involve hard manual labor, and expanded public works, which involve lighter tasks and access to livestock [32]. In alignment with these efforts, Sugira Muryango, an evidence based home visiting intervention, was implemented as a cluster-randomized trial across three districts: Nyanza, Ngoma, and Rubavu, targeting poor households eligible for VUP, with children aged 6–36 months [33, 34]. The intervention delivered home-based coaching sessions aimed at promoting responsive parenting, father engagement, violence prevention, early stimulation, and improved nutrition practices [33, 34]. The evidence from the Sugira Muryango cRCT indicated that the program improved child development outcomes, dietary diversity, caregiver engagement, and reduced harsh discipline [33, 34]. However, it remains unclear whether caregiver household type (single vs. dual caregiver) modified these outcomes.

Given the unique challenges faced by single caregivers, particularly in low-resource settings, this study reanalyzed the Sugira Muryango cRCT sample to assess whether the effects of the intervention on child growth, dietary diversity, and care-seeking behaviors varied by caregiver household type. Evidence of variation could highlight the need for tailored strategies to ensure equitable benefits, whereas a lack of variation may indicate the intervention's potential applicability across diverse household structures.

Methods

Research setting

Rwanda is a landlocked country in central Africa, classified as a low-income country, with a gross domestic product (GDP) of 17.6 billion USD in 2024 [35]. The population was estimated at 13,798,561 in the same year, and the life expectancy at birth was 69.9 years [36]. Rwanda is divided into 30 districts spread across four administrative provinces (South, North, East, and West) and the capital city of Kigali. Each district is subdivided into sectors that are further divided into cells, with the smallest administrative unit being the Umudugudu (village). Sugira Muryango was implemented as a cluster-randomized trial in three districts, Nyanza in the Southern Province, Ngoma in the Eastern Province, and Rubavu in the Western Province, strategically selected for their minimal overlap with other community interventions. The program targeted the poorest households eligible for the VUP, a national social protection initiative that provides direct or conditional cash transfers to very poor households to accelerate poverty reduction and strengthen social safety nets.

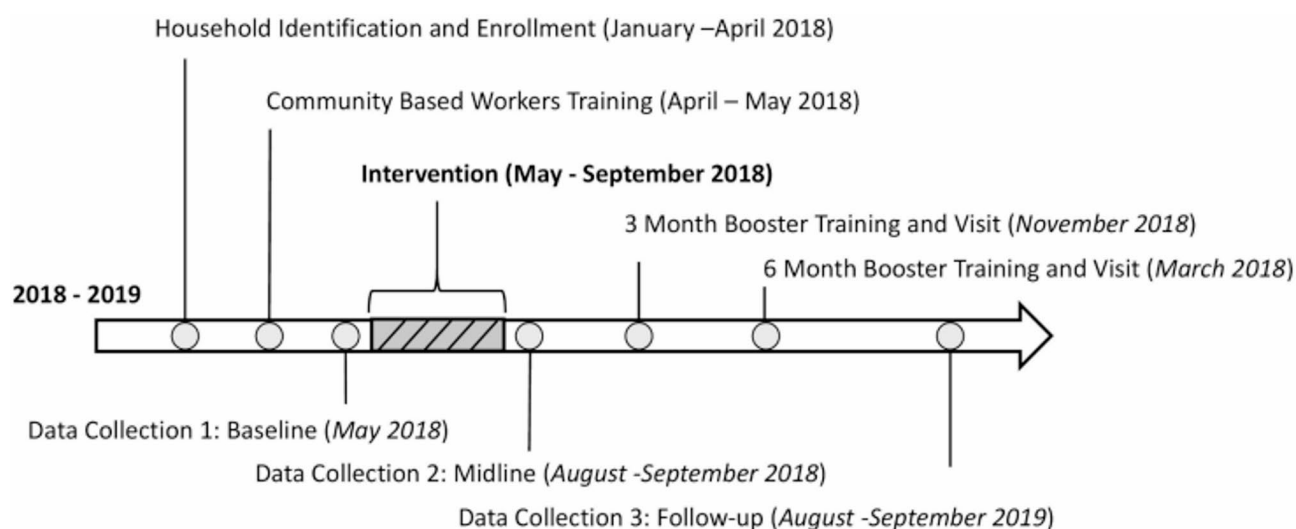
Study design and sampling approach

This study is a secondary analysis of the Sugira Muryango cRCT [34], with data analyzed at baseline and at

12 months after the intervention. Sugira Muryango was implemented using a stratified cluster sampling technique. Clusters were geographically determined to ensure that each cluster contained at least 30 households engaged in classic public works or 10 households engaged in expanded public works. The clusters were spaced as far apart as possible to minimize contamination across study arms. Within each cluster, all households benefiting from Expanded Public Works were sampled, and a random sampling approach was used to select households benefiting from Classic Public Works.

Study period

Enrollment in the Sugira Muryango cRCT program occurred between January and April 2018. Baseline data were collected in May 2018. The intervention was conducted from May to September, 2018 with follow-up assessments occurring three months after the intervention ended and again after 12 months. We analyzed baseline and endline measurements, excluding midline data, to assess changes in response to Sugira Muryango over the entire study period. Further details on the study period and participants recruitment are provided elsewhere [33].



Source: Effect of a home-visiting parenting program to promote early childhood development and prevent violence: A cluster-randomized trial in Rwanda. *BMJ Glob Health*. 2021 Jan 29;6.

Table 1 Participant characteristics at enrollment

	Single	Dual
Child Sex N (%)	498 (51.9%)	461 (48.1%)
Female	244 (49.0)	118 (50.9)
Male	254 (51.0)	114 (49.1)
Child age N (%)		
Below 2 years (6–24 months)	270 (54.2)	268 (58.1)
Above 2 years (above 24 months)	228 (45.8)	193 (41.9)
Children age (months) mean (sd)	22.4 (8.0)	21.9 (8.0)
Caregiver age (year) mean (sd)	32.7 (10.2)	37.1 (10.2)
Caregiver relationship with child N (%)		
Mother	433 (86.9)	241 (52.3)
Father	4 (0.8)	196 (42.5)
Other	61 (12.3)	24 (5.2)
Caregiver education level N (%)		
No school	87 (17.5)	130 (28.2)
Below 6 years primary	264 (53.0)	201 (43.6)
Above 6 years primary	61 (12.2)	79 (17.1)
Secondary/vocational school	86 (17.3)	51 (11.1)
District N (%)		
Nyanza	162 (32.5)	160 (34.7)
Ngoma	200 (40.2)	119 (25.8)
Rubavu	136 (27.3)	182 (39.4)
Children's anthropometry mean (sd)		
WAZ	−1.0 (1.0)	−0.9 (1.0)
HAZ	−2.0 (1.2)	−2.0 (1.1)
HWZ	0.0 (1.0)	0.1 (1.0)
Food group mean (sd)	2.8 (1.2)	2.8 (1.2)
Prevalence of diarrhea N (%)	164 (32.9)	159 (34.5)
Care seeking for diarrhea N (%)	130 (79.3)	119 (74.8)
Prevalence of cough & fever N (%)	237 (54.8)	249 (54.0)
Care seeking for cough & fever N (%)	225 (82.4)	212 (85.1)

Study population and eligibility criteria

Sugira Muryango enrolled households with children aged 6 to 36 months across three districts in Rwanda (Nyanza, Ngoma, and Rubavu). Eligibility criteria included households identified as poor and eligible for the VUP social protection program. At baseline, data were collected from 1,084 children from 1,049 households enrolled in the trial, with 541 households allocated to the intervention group and 508 to the usual care group [33, 34]. Loss to follow-up between baseline and the post-intervention assessment was minimal (< 2.5%), with only three households (0.3%) and six children (0.5%) not completing the post-intervention assessment. Of the assessed sample, 119 children were excluded due to incomplete data, resulting in a final analytic sample of 959 children from both single- and dual-caregiver households. Participant demographics including household structure are summarized in Table 1.

Sugira Muryango intervention

Sugira Muryango was delivered between May and September 2018, with staggered rollout across districts. The home-based intervention aimed to promote responsive caregiving, nutrition, hygiene, and nonviolent parenting through active coaching and psychoeducation, with a focus on engaging both male and female caregivers. Each session included a 15 min guided play activity to support responsive parent-child interaction, aligned with the UNICEF/WHO Care for Child Development package and the Nurturing Care Framework [37]. The intervention consisted of 12 weekly modules, followed by two booster sessions delivered approximately 3 and 6 months after the main intervention period. These booster sessions aimed to reinforce key messages, revisit family goals, and troubleshoot any barriers to sustaining behavior change. Each booster visit lasted approximately one hour and was conducted in person by the trained coaches. Community-based coaches delivered all sessions and received ongoing supervision, peer support, and fidelity monitoring. Beyond parenting support, coaches also helped families access formal and informal resources to address broader needs, including nutrition and conflict resolution. Resource linkage support was primarily provided during the main intervention period, when coaches actively assessed family needs and facilitated connections to formal services (e.g., governmental services) and informal supports (e.g., neighbors, extended families). This support was not formally extended beyond the main intervention period, although families were encouraged to continue engaging with local resources independently [33, 34].

Measures

The questionnaire used in this study was previously described in a published pilot study [38].

Caregiver household type was the type of caregiver, assessed at baseline. Although caregiver household type could change over time, only the baseline status was used in this analysis. Children living in single caregiver households were those cared for by only one adult at home, who could be their mother, father, or another relative such as an aunt or uncle. These single caregivers included individuals who were never married, separated, divorced, widowed, or whose partner was otherwise permanently absent from caregiving responsibilities. In contrast, dual caregiver households included two adults sharing responsibilities for the child's upbringing. This could involve a mother and father, a parent with a step-parent, or two relatives such as an aunt and uncle living together. Additionally, some children lived in intergenerational caregiver households, where grandparents or great-grandparents were the primary caregivers. For this study, intergenerational households were categorized as

single or dual caregiver households based on whether a second adult caregiver was present.

Height for age, weight for age and weight for height

Height-for-Age (HAZ), Weight-for-Age (WAZ), and Weight-for-Height (WHZ) z-scores were used as continuous measures to assess children's growth status over time based on the World Health Organization (WHO) growth standards.

Dietary diversity

Dietary diversity was assessed based on the number of food groups consumed by the child in the past 24 h. Caregivers were asked to recall all foods and beverages their child had consumed during this period [39]. The reported foods were then categorized into seven food groups: [1] cereals and grains [2], roots and tubers [3], legumes and nuts [4], dairy products [5], vitamin A-rich vegetables and fruits [6], other fruits and vegetables, and [7] animal-source foods. Dietary diversity was measured as the total number of distinct food groups consumed.

Parental care seeking behaviors

Care seeking behavior was evaluated by asking caregivers whether they had sought care for three common childhood illnesses: diarrhea, cough, and fever. Affirmative responses were coded as 1, while negative responses were coded as 0.

Ages and stages questionnaire version 3 (ASQ-3)

The ASQ-3 is a caregiver-reported, age-dependent standardized screening tool designed to assess early childhood development across five key domains: communication, gross motor, fine motor, problem solving, and personal-social skills [40, 41]. Each domain includes six items (totaling 30 items per age-specific questionnaire), with questions tailored to the child's age in months to assess whether developmental milestones are met. Caregivers report whether their child performs each behavior "yes," "sometimes," or "not yet," which are scored as 10, 5, or 0, respectively. The tool has been widely validated in multiple contexts, including low-resource settings. In this study, raw domain scores were summed and converted into z-scores for analysis to allow comparability across age groups.

Control variables

child's sex, age, residence, caregiver's education level, age, and relationship to the child, were included in the analysis.

Statistical analysis

We used STATA/SE software version 18.0 for statistical analysis. We conducted analyses including both the intervention and control groups to test if caregiver household type (single vs. dual caregiver) modified the effect of the intervention. Descriptive statistics were performed for

Table 2 Modification effect of caregiver household type on the effects of the Sugira Muryango intervention

	Household x Timepoint interaction		Intervention x Household x Timepoint interaction	
	Coefficient (95% CI)	p-value	Coefficient (95% CI)	p-value
Child development outcomes [#]				
ASQ-3 Gross motor z-score	-0.087 (-0.576,0.401)	0.72	0.106 (-0.570,0.784)	0.75
ASQ-3 Fine motor z-score	-0.036 (-0.368,0.296)	0.83	0.129 (-0.331,0.589)	0.58
ASQ-3 Communication z-score	0.125 (-0.240,0.491)	0.50	-0.272 (-0.779,0.235)	0.29
ASQ-3 Problem solving z-score	0.084 (-0.257,0.425)	0.62	0.101 (-0.371,0.574)	0.67
ASQ-3 Social emotional z-score	0.112 (-0.221,0.446)	0.50	0.090 (-0.372,0.553)	0.70
Child growth outcomes [#]				
Height for age z-score (HAZ)	-0.056 (-0.196,0.083)	0.43	0.013 (-0.180,0.206)	0.89
Weight for age z-score (WAZ)	-0.090 (-0.204,0.023)	0.11	0.081 (-0.076,0.239)	0.31
Weight for height z-score (WHZ)	-0.040 (-0.200,0.280)	0.26	0.128 (-0.083,0.341)	0.23
Child nutrition and health				
Dietary diversity [#]	-0.040 (-0.351,0.270)	0.79	0.255 (-0.174,0.686)	0.24
Diarrhea prevalence *	-0.193 (-0.857,0.470)	0.56	0.075 (-0.837,0.987)	0.87
Diarrhea care seeking *	-1.514 (-3.615,0.589)	0.15	2.025 (-0.920,4.972)	0.17
Cough and fever prevalence *	0.350 (-0.220,0.921)	0.22	-0.373 (-1.167,0.420)	0.35
Cough and fever care seeking *	0.684 (-1.656,0.287)	0.16	1.215 (-0.133,2.564)	0.07

Adjusted for child sex, child age, caregiver's relationship with the child, caregiver age, and caregiver education

Intervention: 0 = control group, 1 = Sugira Muryango intervention

Household type: 0 = single caregiver households, 1 = dual caregiver households

Timepoint: 0 = baseline, 1 = endline

[#] LMM regression coefficients represent mean differences

*Log-odds coefficients are from mixed-effects logistic regression models

Predicted Growth and Dietary Outcomes Over Time by Household type

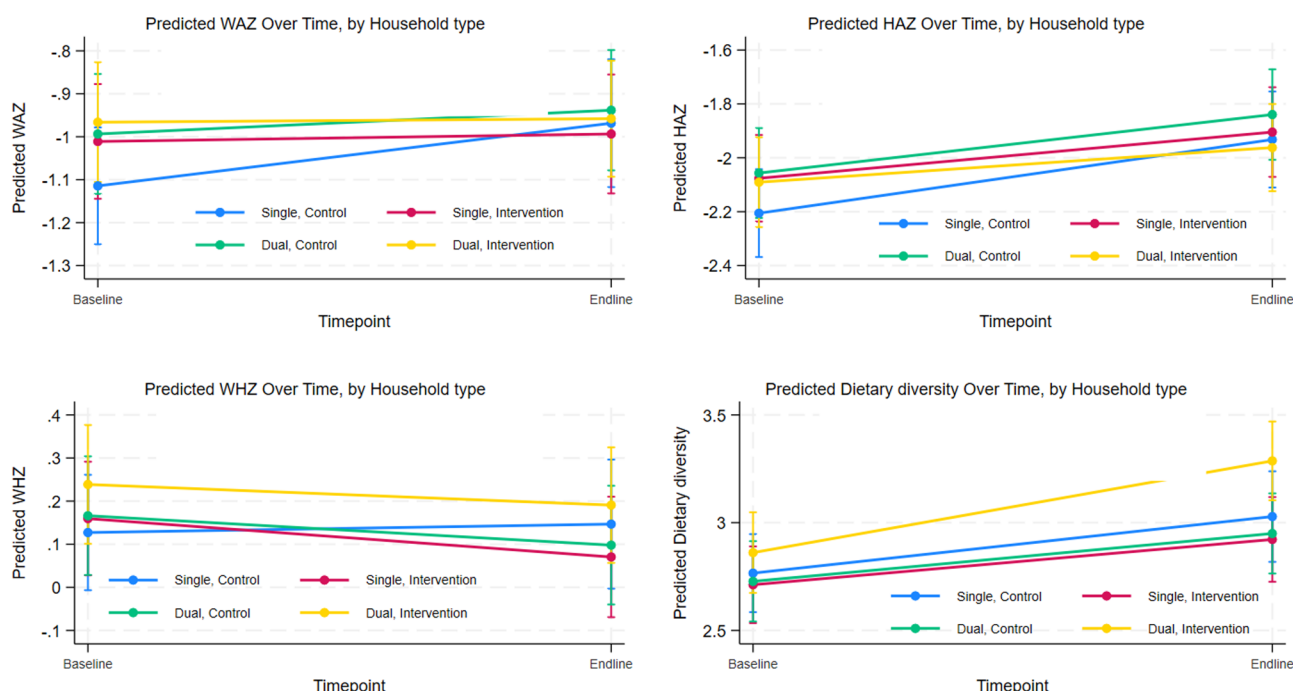


Fig. 1 Predicted growth and dietary outcomes values

demographic and socioeconomic variables. Categorical variables were summarized using frequencies and percentages, while continuous variables, which were approximately normally distributed, are presented as means and standard deviations. We used Linear mixed-effects models (LMMs) to account for the hierarchical structure of the data, with children nested within households and clusters. The analytic dataset included one child-caregiver dyad per household. Random intercepts for households accounted for repeated measurements within each dyad over time (baseline and endline), and random intercepts for clusters addressed potential intra-cluster correlation, as clusters were the unit of randomization in the original cRCT. The LMMs included fixed effects for timepoint (baseline vs. endline), treatment group (intervention vs. control), caregiver household type (single vs. dual), and their three-way interaction (timepoint $\times$ treatment $\times$ caregiver household type) to assess whether changes in outcomes over time differed by caregiver household type. For binary outcomes, such as care seeking for diarrhea, cough, and fever, we used mixed-effects logistic regression models. To control for potential confounders, we adjusted for child-level covariates (age and sex) and caregiver-level covariates (caregiver age, education level, and relationship to the child). We present LMM regression coefficients with their corresponding 95% confidence intervals (CIs) for continuous outcomes.

For binary outcomes, we report adjusted log-odds coefficients with 95% CIs derived from mixed-effects logistic regression models.

Results

Sample description at baseline

A total of 1,084 children from 1,049 households were initially enrolled. Children were assessed at baseline, and follow-up data were collected 12 months after the intervention. Due to missing data, the final analytic sample included 959 children with complete data. Of these, 498 (51.9%) were from single caregiver households and 461 (48.1%) from dual caregiver households. At baseline, WAZ, HAZ, WHZ scores, and dietary diversity were similar across caregiver household types. The prevalence of diarrhea was 32.9% in single caregiver households compared to 34.5% in dual caregiver households. The prevalence of cough and fever was comparable across both groups. Care-seeking for diarrhea was higher in single-caregiver households (81%) than in dual-caregiver households (75%) at baseline (Table 1).

Results from linear mixed models

We conducted LMMs, accounting for household and cluster levels, to explore differences in outcomes between single and dual caregiver households.

Predicted Developmental and Care seeking Outcomes Over Time by Group

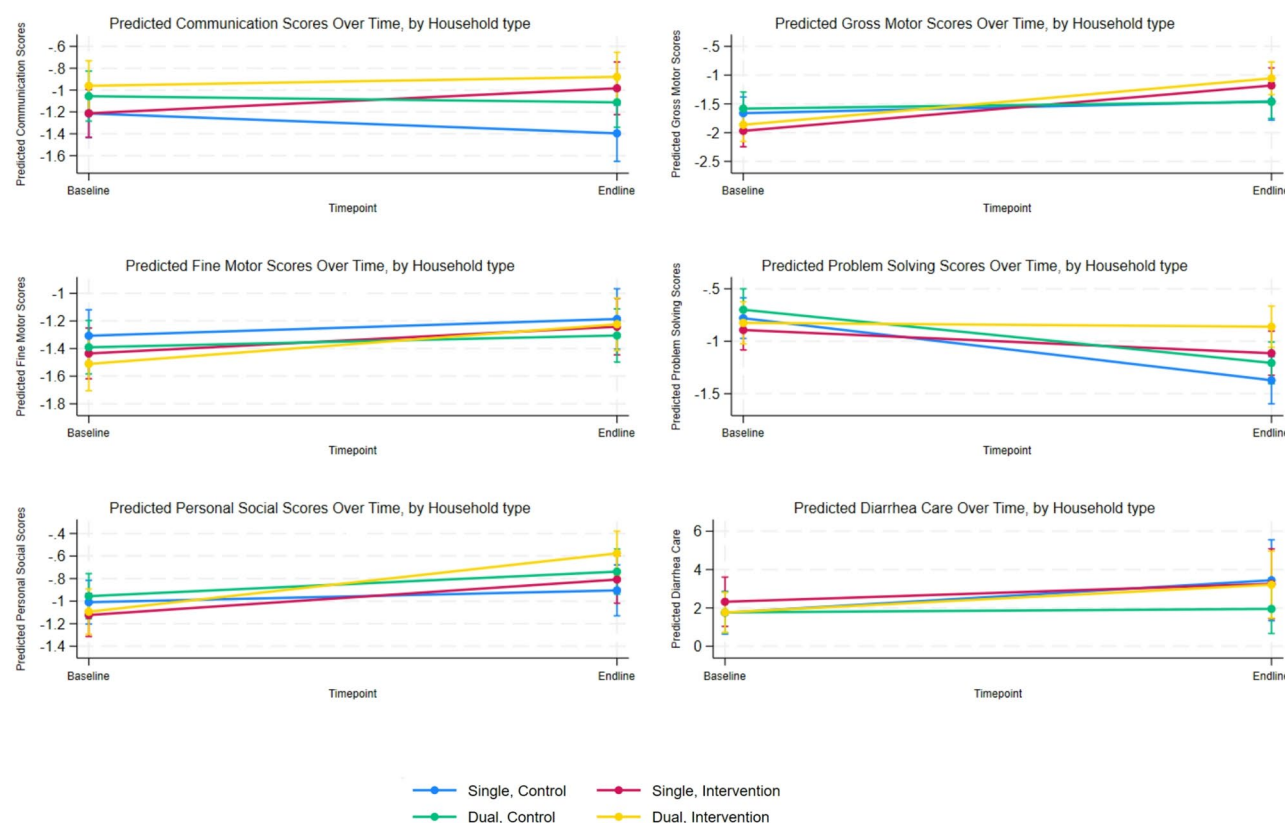


Fig. 2 Predicted ASQ-3 scores values

WAZ, HAZ and WHZ

There was no evidence of differential changes in WAZ over time by caregiver household type in response to the intervention ($\beta = 0.081$, 95% CI: -0.077 to 0.239 , $p = 0.313$) (Table 2). At endline, predicted WAZ was similar across caregiver household types within both the control group (-0.97 for single caregivers vs. -0.94 for dual caregivers) and the intervention group (-0.99 for single caregivers vs. -0.96 for dual caregivers), indicating no evidence of differential effects by caregiver household type (Fig. 1; Table S1).

For HAZ, changes over time did not differ between the intervention and control groups ($\beta = 0.013$, 95% CI: -0.180 to 0.207 , $p = 0.893$) (Table 2). Predicted margins indicated slight improvements in HAZ from baseline to endline across all groups. Among children in dual-caregiver households in the intervention group, HAZ improved from -2.09 at baseline to -1.96 at endline, while in single-caregiver households it improved from -2.08 to -1.90 , suggesting similar modest gains irrespective of caregiver household type or intervention status (Fig. 1; Table S1).

Similarly, there was no evidence of a differential intervention effect over time by caregiver household type on WHZ ($\beta = 0.129$, 95% CI: -0.083 to 0.341 , $p = 0.234$) (Table 2). Predicted values showed WHZ remained

relatively stable across groups. In dual-caregiver households within the intervention group, WHZ changed slightly from 0.24 at baseline to 0.19 at endline, while in single-caregiver households it changed from 0.16 to 0.07 , reflecting small, non-significant declines in both groups over time (Fig. 1; Table S1).

Dietary diversity

Changes in dietary diversity did not differ by caregiver household type in response to the intervention over time ($\beta = 0.256$, 95% CI: -0.175 to 0.686 , $p = 0.244$) (Table 2). Predicted values showed increases in the number of food groups consumed from baseline to endline across all groups. In dual-caregiver households within the intervention group, the predicted count increased from 2.86 at baseline to 3.29 at endline, while in single-caregiver households it increased from 2.71 to 2.92 , reflecting slight improvements in dietary diversity over time regardless of caregiver household type (Fig. 1; Table S1).

Early childhood development

There was no statistically significant modification effect of caregiver household type for any ASQ domain, indicating no differential intervention effects over time between

children from single- and dual-caregiver households (Table 2). For the individual domains, the three-way interaction terms were not significant: Communication ($\beta = -0.272$, 95% CI: -0.779 to 0.235 , $p = 0.293$), Gross Motor ($\beta = 0.107$, 95% CI: -0.571 to 0.785 , $p = 0.757$), Fine Motor ($\beta = 0.129$, 95% CI: -0.331 to 0.590 , $p = 0.582$), Problem-Solving ($\beta = 0.101$, 95% CI: -0.372 to 0.574 , $p = 0.675$), and Personal-Social ($\beta = 0.090$, 95% CI: -0.370 to 0.550 , $p = 0.700$).

Predicted margins showed modest and comparable improvements across most ASQ developmental domains in response to the intervention, irrespective of caregiver household type. Among children in the intervention group, changes were as follow: For communication, dual-caregiver children improved from -0.96 at baseline to -0.88 at endline, while single-caregiver children improved from -1.21 to -0.98 . In the gross motor domain, both groups demonstrated similar improvements, with dual-caregiver children increasing from -1.86 to -1.06 and single-caregiver children from -1.97 to -1.18 . For fine motor skills, slight improvements were noted across groups; dual-caregiver children improved from -1.51 to -1.22 , and single-caregiver children from -1.44 to -1.24 . In problem-solving, predicted margins indicated small but comparable changes, with dual-caregiver children changing from -0.82 to -0.86 and single-caregiver children from -0.89 to -1.11 . Lastly, in the personal-social domain, improvements were observed in both groups, with dual-caregiver children increasing from -1.09 to -0.58 and single-caregiver children from -1.12 to -0.81 (Fig. 2; Table S1).

Care-seeking behavior for diarrhea, fever and cough

We observed no statistically significant modification effect of caregiver household type on care-seeking outcomes over time. For diarrhea care-seeking, the three-way interaction coefficient was 2.03 (95% CI: -0.92 to 4.97 , $p = 0.178$), indicating no evidence of a differential intervention effect by caregiver household type. For fever/cough care-seeking, the three-way interaction coefficient was 1.22 (95% CI: -0.13 to 2.56 , $p = 0.077$), similarly indicating no significant effect modification (Table 2). For single-caregiver households, the predicted log-odds of care-seeking for diarrhea increased from 1.75 at baseline to 3.44 at endline in the control group, and from 2.32 to 3.26 in the intervention group. Among dual-caregiver households, the log-odds increased from 1.77 to 1.95 in the control group, and from 1.76 to 3.21 in the intervention group (Fig. 2; Table S1).

Discussion

This study assessed whether the caregiver household type modified the effects of Sugira Muryango, a home-visiting program in Rwanda, on child growth, development,

dietary diversity, and care-seeking. The findings suggest that caregiver household type did not modify these outcomes. No significant differences were observed in changes in child growth (WAZ, HAZ, WHZ), dietary diversity, developmental outcomes across ASQ domains, or care-seeking behaviors for diarrhea, cough, and fever between single and dual caregiver households.

Our results indicate that the Sugira Muryango intervention produced similar outcomes in children's growth, development, dietary diversity, and care-seeking behavior among caregivers in both single and dual caregiver households, with no significant differences between the two groups. Studies have identified single caregiver household as a vulnerability factor for poorer child health outcomes [42–44]. For instance, research in sub-Saharan Africa has shown that children in single-parent households, particularly those led by mothers, often experience worse nutritional and developmental outcomes [42–44] due to financial and social constraints [22, 45, 46]. However, well-designed, inclusive interventions may help prevent the widening of poverty-related disparities affecting single-caregiver households [33, 47, 48]. The results of Sugira Muryango may have been driven by key program components, including psychoeducation on child development and nutrition, active stimulation and responsive parenting coaching, problem-solving skills, and strategies for positive parenting and coping with parental stress [33, 34]. These elements likely strengthened parenting practices, fostered nurturing interactions, and enhanced family resilience, even in single caregiver households facing heightened caregiving and economic burdens, such as limited income and fewer employment opportunities, challenges that are particularly relevant in the context of Rwanda [49].

The intervention outcomes for single and dual caregiver households were similar, but it is important to note that these groups may have contextual differences. Single caregivers in Rwanda and other low-resource settings face unique challenges such as economic insecurity, time constraints, and limited childcare support [16–19]. While Rwanda offers some social protection for impoverished households, child protection services, and support for children with disabilities [50–52], there are no specific policies targeting single-parent households. Many single caregivers rely on informal support from extended family members, particularly grandmothers and older siblings, though this support is inconsistent. This context highlights the need to consider the specific challenges faced by single caregivers when designing and implementing interventions. The Sugira Muryango intervention aimed to connect caregivers to both formal and informal support networks, but a deeper understanding of these challenges is important for improving the design of similar interventions in different settings.

Diverse caregiving arrangements likely influence family dynamics and, consequently, child outcomes. Single caregiver households, may face specific challenges that affect the quality and consistency of care provided [16–19]. In contrast, dual caregiver households benefit from shared responsibilities between two adults, potentially offering more balanced support for the child's needs. It is also important to acknowledge that intergenerational caregiving involves distinct family dynamics compared to parental caregiving structures. In households where grandparents or great-grandparents serve as primary caregivers, caregiving styles, available resources, and intergenerational relationships differ from those in parent-led households [53, 54]. These families may face unique challenges and strengths, such as culturally shaped caregiving norms, varying health and energy levels of older caregivers, and different support networks [55]. Such complex dynamics emphasize the importance of understanding the varied contexts in which caregiving occurs, as they can shape intervention effectiveness and influence pathways of child development.

The Sugira Muryango intervention did not have a significant impact on children's physical growth, and this lack of effect was consistent across caregiver household types. The high proportion (48%) of stunted children at baseline, in our sample [33], suggests that many children had already experienced early growth deficits that may be difficult to reverse [56]. Evidence from LMICs shows that postnatal linear growth faltering typically occurs from birth to three years of age, resulting in declines in mean HAZ and a downward shift in the overall distribution for nearly the entire population under three years, not just vulnerable subgroups [57]. Once established, stunting becomes increasingly resistant to change, especially after the first two years of life when interventions have the greatest potential to improve growth outcomes [56]. Younger children are more likely to benefit from nutrition and health interventions due to their potential for rapid catch-up growth when provided with adequate care. However, as children age, the window of opportunity for recovery narrows [56, 58], and older children who have experienced prolonged periods of stunted growth prior to intervention are unlikely to catch up [56, 59]. Furthermore, while children on normal growth trajectories tend to maintain their z-scores over time with minimal variation [60, 61], growth trajectories in this sample may have been constrained by chronic undernutrition and repeated exposure to infections [62]. This is particularly concerning given the elevated prevalence of illnesses such as diarrhea, cough, and fever, compounded by prolonged social and environmental adversity, which further undermine children's growth and overall health.

We observed a significant increase in the number of food groups consumed by children from baseline to

endline across all groups. This finding aligns with previous research indicating that community-based programs targeting parenting and nutritional education can improve children's dietary diversity [63, 64]. Notably, the educational content delivered to caregivers as part of the intervention may have contributed to these improvements, as it equipped them with knowledge and skills to diversify meals using available resources. This is consistent with existing literature highlighting that educated caregivers are better able to understand the importance of dietary diversity and apply this knowledge in practice [65, 66]. These findings emphasise the value of integrating nutrition education into community-based interventions to enhance child feeding practices and dietary outcomes [67].

Sugira Muryango led to improvements in children's development across ASQ domains, irrespective of caregiver household type. This stresses the critical role that caregiver training and skill-building delivered through the intervention play in promoting child development by increasing caregivers' awareness of optimal parenting practices, improving their ability to access health and educational resources, and enhancing their capacity to engage in stimulating interactions with their children, all of which contribute to better developmental outcomes [68]. Indeed, interventions that emphasize caregiver skill-building have consistently demonstrated positive effects on children's cognitive, motor, and social development [69, 70].

Strength of the study This study addressed a significant gap in the literature by exploring the under examined population of single caregivers, a vulnerable group often underrepresented in research despite being at higher risk for poor child health outcomes in resource-limited settings. Additionally, by leveraging data from the Sugira Muryango intervention, an evidence-based program implemented under real-world conditions, the study offers valuable insights into the practical design of community-based interventions.

Limitations of the study This study has limitations that should be considered when interpreting the findings. First, although we adjusted for confounders, residual confounding from factors such as caregiver mental health and intra-household food distribution may have influenced child growth and developmental outcomes. Future research should incorporate these variables to better elucidate their role in intervention responsiveness. Second, the study was conducted within the specific context of the Sugira Muryango home-visiting program in Rwanda; therefore, the generalizability of the results to other settings with different social, economic, and health system environments may be limited. Replication of similar interventions in diverse contexts is needed to assess broader applicability. Third, variation in the fidelity of intervention delivery across households and clusters was

not fully controlled for in the analysis, which may have impacted the outcomes.

Conclusion

Our findings indicate that the Sugira Muryango intervention produced similar outcomes in child growth, development, food diversity, and care-seeking behavior across both single and dual caregiver households. Its inclusive design and ability to deliver equitable benefits across diverse family settings in low-resource contexts highlight its potential as a scalable model for community-based interventions. Key features, such as the flexible, home-based format that reduced travel burdens and allowed coaches to tailor support to each household's unique needs, and modules on stress management, responsive caregiving, and problem-solving strategies were particularly supportive of single caregivers, enhancing the intervention's cultural and contextual relevance. These findings have important implications for implementing parenting and family-strengthening programs in resource-limited settings. The absence of significant differences between single and dual caregiver households is particularly noteworthy, given the often-cited vulnerabilities of single caregivers, especially female-headed households, in achieving positive child health outcomes.

Our findings suggest potential areas to strengthen the Sugira Muryango intervention, particularly by placing greater emphasis on early intervention to improve child growth outcomes. Since growth faltering becomes increasingly difficult to reverse after the age of two, it is critical to intensify efforts to reach and support caregivers and children during pregnancy and the first year of life. Early engagement may provide a more effective window for promoting optimal nutrition and development, thereby reducing the risk of chronic undernutrition and stunting. Additionally, although caregiver household type did not significantly modify the intervention's effects on child growth and development, there remains value in tailoring intervention strategies to address unique barriers faced by different household types. Customizing approaches to reflect the specific needs, resources, and challenges of single-caregiver versus dual-caregiver households may enhance caregiver engagement and program responsiveness, ultimately improving child health and developmental outcomes.

Abbreviations

SM	Sugira Muryango
NIH	National Institute for Health
WHO	World Health Organization

Supplementary Information

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Supplementary Material 1.

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Authors' contributions

UJ performed the literature review, data analysis, and interpretation of the results and contributed to manuscript preparation. VS and TSB contributed the conception and design of the primary study. JB, VS, MP, and TSB participated in the interpretation of results, critical revision of the manuscript, and supervision of the study. All authors have reviewed the results and approved the final version of the manuscript.

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Data availability

The authors declare that the data supporting the findings of this study are available upon reasonable request from the Principal Investigator, Dr. Theresa Betancourt. Additionally, the data are publicly accessible through the World Bank website. For access, please visit: <https://microdata.worldbank.org/index.php/catalog/6107>.

Declarations

Ethics approval and consent to participate

This study was conducted in compliance with the ethical principles outlined in the Declaration of Helsinki. The primary study received approval from the Institutional Review Boards of Harvard University, Chan School of Public Health, and Boston College, as well as from the Rwanda National Ethics Committee, National Committee for Science and Technology, and the National Institute of Statistics of Rwanda. All primary caregivers provided written informed consent for their participation and that of their children.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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